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Scandinavian Journal of Medicine & Science in Sports

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ORIGINAL ARTICLE OPEN ACCESS

How Humans Run Faster: The Neuromechanical Contributions of Functional Muscle Groups to Running at Different Speeds

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Received: 11 December 2023 | **Revised:** 30 April 2024 | **Accepted:** 21 June 2024

Funding: This work was in part financially supported by the English Institute of Sport.

Keywords: EMG | inverse dynamics | kinematics | kinetics | running

ABSTRACT

How the neuromechanics of the lower limb functional muscle groups change with running speed remains to be fully elucidated, with implications for our understanding of human locomotion, conditioning, and injury prevention. This study compared the neuromechanics (ground reaction and joint kinetics, kinematics and muscle activity) of middle-distance athletes running on an instrumented treadmill at six wide-ranging speeds (2.78–8.33 m·s^{-1}). Ground reaction forces and kinematics were analyzed using inverse dynamics to calculate flexor and extensor joint torques, and positive and negative work done by these torques. Contributions of each functional muscle group to the total positive and negative work done by the limb during stance, swing, and the whole stride were quantified. During stance, the ankle plantar flexors were the major energy generator and absorber (>60%) at all speeds, but their contribution to whole stride energy generation and absorption declined with speed. Positive work by the hip extensors rose superlinearly with speed during stance (3-fold) and especially during swing (12-fold), becoming the biggest energy generator across the whole stride at >5 m·s^{-1}. Knee flexor and extensor negative work also rose superlinearly with speed during swing, with the knee flexors becoming the greatest energy absorber over the whole stride at >7.22 m·s^{-1}. Across speeds, plantar flexor peak moment and positive work accounted for 97% and 96% of the variance in step length, and swing hip extension peak moment and positive work accounted for 98% and 99% of the variance in step frequency. There were pronounced speed, phase (stance/swing), and work (positive/negative) dependent contributions of the different functional muscle groups during running, with extensive implications for conditioning and injury prevention.

1 | Introduction

Running is a fundamental means of human locomotion, an extremely popular physical activity, and the basis of many sports. Running can be performed at a wide range of speeds that have different physiological and biomechanical requirements [1]. Understanding lower limb neuromechanics at different running speeds may reveal the contribution and importance of specific muscle groups to running at each speed, which may inform preparation for athletic events and prevention of injury. However, knowledge of the neuromechanics of running at different speeds from jogging through to sprinting remains incomplete.

Running speed appears to increase primarily due to increased step length (SL) at low speeds, and increased step frequency (SF)

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at high speeds [2, 3]. However, the exact contributions of these factors have not been quantified. Furthermore, these speed-dependent changes in stride parameters would be expected to require modified kinematics and kinetics of the lower limb joints and muscles that in turn depend on changes in muscle activation. Regarding kinematics, faster running is known to be achieved with higher knee and hip joint flexion-extension ranges of motion [4] and increased joint and segment angular velocities [5, 6]. However, the nature of these changes, particularly in lower limb joint angular velocities over a wide range of running speeds has not been comprehensively analyzed.

In contrast, joint kinetics and particularly peak joint torques/powers during running have received more attention. During the stance phase of running, peak ankle moments increase with running speed, but there is contradictory evidence for peak knee joint moments/powers increasing [5, 7, 8], remaining unchanged [9, 10] or decreasing [11]. These contradictory findings may be because previous studies analyzed only one [7–11] or three [5] contacts per participant, or assessed only a small number and/or a limited range of running speeds. Analyzing multiple strides per trial across a wide range of speeds would be expected to generate more reliable data.

Importantly, the mechanical work done about the joints (i.e., the integral of the power–time curve), rather than instantaneous moments or powers, may better inform how the contributions of the different lower limb joints/muscles change with running speed. This approach has only been taken by a few studies [5, 7, 9, 12–14]. In the most thorough studies to date, Schache and colleagues [9, 13] found that the positive and negative work done at the ankle joint during stance was relatively high and increasing at slow speeds before plateauing at higher speeds, whereas the hip joint positive work and knee joint negative work during swing increased quadratically with speed. However, this study employed an analysis that did not distinguish between the contributions of opposing muscular joint moments about each joint (i.e., flexor and extensor moments by functional muscle groups) to the positive and negative work done at each joint during running. In contrast, by analyzing the joint kinetics on the basis of individual direction-specific moments about a joint (i.e., flexion and extension moment), it is possible to assess the work done by functional muscle groups consisting of all muscle-tendon units contributing to the moment [5, 15].

At present, how the absolute and proportional (i.e., relative to overall work done by the lower limb) work contributions of different functional muscle groups change with speed during both stance and swing phases, and particularly during a whole stride cycle (i.e., summation of stance and swing) remains unknown. On the basis of hip joint moments, it has been suggested that the hip extensors make an increasing contribution to both swing and stance phase positive work as running speed increases [5, 13, 16] and its possible that this could exceed the positive work done by the plantar flexors at high speeds. The negative work done by the knee flexors (during swing) appears to increase sharply during fast running [17] and could potentially exceed the energy absorbed by the plantar flexors (during stance), which appears to plateau at high speeds [13]. However, a thorough analysis of the positive and negative work done by each functional muscle group in absolute and proportional terms is needed to test these possibilities.

Using musculoskeletal modeling, Dorn, Schache, and Pandy [3] estimated the kinetic contributions of individual leg muscles at different running speeds and attempted to link these parameters to the changes in stride parameters. They found the plantar flexors to be the major contributor to high ground reaction forces/impulse, and thus, they inferred stride length, while the hip muscles were critical for lower limb accelerations, and thus, they inferred stride rate. However, these inferences may not be entirely justified, as for example stride length increased up to maximum speed, despite vertical impulse decreasing to a minimum at this point. As no explicit relationships between the actions of functional muscle groups and stride parameters have been determined, direct assessment of the relationship between plantar flexor kinetics and stride length and hip extensor kinetics and stride rate appear warranted.

Muscle activation drives the generation of muscle moments, work, and ultimately forward propulsion via ground reaction forces (GRF). Increases in running speed have been linked to increased muscle activations of all the lower limb muscle groups [18–21]. However, several studies specifically focused on a limited number of muscles (e.g., hamstrings or gluteals [20, 22]), and no studies have assessed muscle activity alongside functional muscle group (or joint) specific kinetics during running at different speeds. Knowledge of how muscle activity and functional muscle group kinetics change may give a more thorough understanding of the muscular demands of running at different speeds. Overall, a full analysis of running neuromechanics, combining kinematics, kinetics, and muscle activation for the whole of multiple gait cycles, over a wide range of running speeds has not been completed. Therefore, the aim of this study was to compare the kinematics, kinetics, and muscular activity of the lower limbs of athletes running at six different wide-ranging running speeds (2.78–8.33 $m \cdot s^{-1}$). It was hypothesized that during a whole stride cycle: (i) energy generation (positive work) at slow running speeds would be dominated by the ankle plantar flexors, but this would plateau with increasing speed, and be overtaken by the hip extensors as the largest energy generator at high speeds; (ii) the primary energy absorber (negative work) would be the ankle plantar flexors at low speeds, but this would be exceeded by the knee flexors at high speeds; and (iii) plantar flexor stance kinetics would be positively related to SL and hip extensor swing kinetics would be positively related to SF over the range of running speeds.

2 | Methods

2.1 | Participants

Eleven male middle-distance runners (age: 21.7 ± 3.2 years, height: 1.76 ± 0.04 m, weight: 64.4 ± 5.2 kg) took part in this study. Middle-distance runners were recruited since they train at, and are familiar with, a wide range of running speeds [23], rather than sprinters or long-distance runners who may be accustomed to running effectively at a smaller range of speeds.

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Only male participants were considered in order to analyze a homogenous group with relatively similar maximum running speeds, stature, and body composition. Participants had a season's best middle-distance (800 or 1500 m) performance of 806 ± 146 World Athletics points which is equivalent to a 1500 m time of 4:04 min:s. Healthy and injury free participants provided informed consent before participating in this study which was approved by the Loughborough University Ethics Approvals (Human Participants) subcommittee.

2.2 | Experimental Design

Participants attended the laboratory on two occasions, firstly for a familiarization session and secondly the main session of neuromechanical measurements at different running speeds. The sessions were at least 2 days apart and performed on a level instrumented treadmill (3DI, Treadmetrix, Utah, USA). During the familiarization session, participants extensively practiced mounting and dismounting the moving treadmill at each of the speeds to be assessed in the main session, until the participants were comfortable with the procedure at all speeds. Additionally, 45 anthropometric measurements comprising lengths, widths, and perimeters were taken, from which subject-specific inertia parameters were calculated [24]. During the main session, following placement of EMG sensors and reflective markers, participants performed a 5-min warm-up on the treadmill at a self-selected speed before performing three maximum counter movement jumps (CMJs) for EMG normalization. Participants then completed a running protocol on the treadmill at the following six speeds: 2.78 $m \cdot s^{-1}$ (10 $km \cdot h^{-1}$), 3.89 $m \cdot s^{-1}$ (14 $km \cdot h^{-1}$), 5.00 $m \cdot s^{-1}$ (18 $km \cdot h^{-1}$), 6.11 $m \cdot s^{-1}$ (22 $km \cdot h^{-1}$), 7.22 $m \cdot s^{-1}$ (26 $km \cdot h^{-1}$), and 8.33 $m \cdot s^{-1}$ (30 $km \cdot h^{-1}$). Participants mounted the treadmill at full speed and ran for 4 min at the first two speeds, 45, 35, and 20 s, respectively, for the following three speeds, and for at least six strides at 8.33 $m \cdot s^{-1}$. There was a 1-min break between speeds up to 7.22 $m \cdot s^{-1}$ and 2 min after 7.22 $m \cdot s^{-1}$. Data collection involved recording muscle activity and kinetics (both sampled at 2000 Hz), and kinematics (sampled at 250 Hz) synchronously through Vicon Nexus (Vicon, Oxford), with eight strides analyzed per speed up to 7.22 $m \cdot s^{-1}$ and six strides at 8.33 $m \cdot s^{-1}$. Data collection started after 3 min of running at the first two speeds and 25, 15, and 5 s, respectively, at subsequent speeds to ensure participants were in an even paced, habitual running gait after mounting the treadmill. For the fastest speed, data collection had to start immediately after the participant mounted the treadmill to allow at least six strides to be captured.

2.3 | Motion Capture and Kinetics

Eighteen infrared cameras were positioned around the treadmill (Vicon, Oxford), and the system was calibrated with a 5-marker reflective wand. The wand was placed in the middle of the treadmill to define the origin and global coordinate system. Thirty-nine reflective markers were attached to the participant on specific bony landmarks. Four markers placed on the treadmill defined a reference plane of the treadmill surface and the location of the force plate relative to the global coordinate system. Prior to the first running speed, a static trial was collected while participants were standing upright, feet shoulder width apart, facing forward with their arms outstretched by their sides, palms facing forward. Medio-lateral, anterior–posterior and vertical ground reaction forces, center of pressure (COP), and free moment during running were recorded using four calibrated force transducers (MC3A, AMTI, Watertown, USA) embedded within a force plate in the treadmill.

2.4 | Counter Movement Jump

Three maximum counter movement jumps (CMJs) were performed, with a 1-min break between jumps. Participants stood on a force plate embedded in the floor with feet shoulder width apart and maintained their hands on their hips throughout the jump. Participants were instructed to do a maximum CMJ by squatting down and then jumping up as high as possible.

In pilot work, we compared peak EMG amplitudes (100 ms epoch, see below) during the CMJ (EMG_{CMJ}) to those measured at isometric maximum voluntary force (EMG_{iMVF}) during separate isometric maximum voluntary contractions (MVCs) of the plantar flexors, knee extensors, knee flexors, and hip extensors that we have previously used for EMG normalization [15]. We found $EMG_{CMJ} > EMG_{iMVF}$ for six muscles (RF, VL, VM, LG, S, and GMax) and similar for two muscles (LH and MH), in agreement with previous research [25]. Therefore, in this experiment we used the CMJ as the reference task for the EMG normalization due to its time efficiency and equivalent or higher EMG amplitudes than isometric MVCs.

2.5 | Electromyography

sEMG was recorded from the following eight muscles of both legs (i.e., 16 in all) with wireless sensors (Trigno Wireless EMG; Delsys, Natick, USA): gluteus maximus (GMax), rectus femoris (RF), vastus lateralis (VL), vastus medialis (VM), lateral hamstrings (LH), medial hamstrings (MH), gastrocnemius lateralis (LG), and soleus (S). Segment lengths for thigh and shank were defined as the distance between the greater trochanter and lateral epicondyle of the knee and between the lateral epicondyle of the knee and lateral malleolus of the ankle, respectively. Muscles were palpated and outlined to allow the sensor to be positioned over the center (medio-lateral) of the superficial muscle belly, as well as at specific distances along the segment: 55%, 60%, and 35% of thigh length from superior border of the patella for RF, VL, and VM, respectively; 45% of thigh length from popliteal fossa for LH and MH; 65% of shank length from the medial knee joint space for S, and 85% of shank length from calcaneus for LG. For GMax, 50% of the distance between the second sacral vertebrae to the greater trochanter was used. These locations were used to avoid the innervation zones and to maximize signal fidelity [26]. The participant's skin at these sites was shaved, abraded, and cleaned with alcohol before the sensors were attached with adhesive interfaces and secured with tape. The sEMG signals were amplified and filtered at source (×300; 20–450 Hz bandwidth), before further amplification (overall effective gain ×909) and time aligning with kinetic and kinematic data.

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2.6 | Data Processing

Markers were labeled, and gaps were filled using rigid body fill in Nexus2.5 (Vicon, Oxford). Marker and force plate data were low-pass filtered with a second order Butterworth filter with a cutoff frequency of 19 Hz based on residual analysis from lower body markers [27]. Using the same cut-off frequency for motion and force data has previously been shown to provide more accurate joint torques using inverse dynamics [28]. A 15-segment rigid-body model was built in Visual3D, segments were linked by ball joints allowing three rotational, but no translational, degrees of freedom, and joint angle time histories were calculated by matching the model to marker data using inverse kinematics. Initial contact (IC) and toe-off (TO) were identified when vertical GRF crossed a threshold of 40 N. Strides were defined as successive initial contacts of the same leg, and a step was defined from initial contact of the ipsilateral to the contralateral leg.

SF, ground contact time (GCT; from IC to TO of the same leg), flight (from TO to IC of the contralateral leg), swing times (from TO to IC of the same leg), and ground contact distance (GCD; forward movement of the COM during stance) were calculated. SL was derived by dividing treadmill belt speed by SF based on the following relationship between speed and SL and SF:

$$\text{Speed} = \text{SL} \cdot \text{SF} \tag{1}$$

In order to assess the contribution of the changes in SL and SF to increases in speed, percentage changes in all three variables (speed, SL, and SF) were calculated for each increment in speed (i.e., relative to the preceding speed). When expressed as decimals relative to unity, the percentage change in running speed (ΔSpeed) due to percentage changes in SL (ΔSL) and SF (ΔSF) is given by the following equation:

$$1 + \Delta\text{Speed} = (1 + \Delta\text{SL})(1 + \Delta\text{SF}) = 1 + \Delta\text{SL} + \Delta\text{SF} + \Delta\text{SL} \cdot \Delta\text{SF} \tag{2}$$

The right-hand term allows the relative contributions of changes in SL, SF, and the product of these two changes ($\Delta\text{SL} \cdot \Delta\text{SF}$) to the change in speed to be quantified.

Distinctive sagittal plane lower limb joint angles, such as peak joint angles and angles at initial contact and toe-off, were determined during stance and swing. Ankle dorsiflexion, knee extension, and hip flexion were defined as positive. Peak knee and hip angular velocities were calculated during swing about the flexion-extension axis. Lower limb joint moments were calculated via inverse dynamics. Joint power was derived by multiplying each of the sagittal plane joint moments by the sagittal plane joint angular velocity. Power–time histories in the direction of each sagittal plane moment (flexion/extension) about each joint (ankle, knee, and hip) were then numerically integrated during stance and swing phases separately to calculate positive and negative work done by this moment. Additionally, mechanical work (both positive and negative) of each muscle group over a whole stride cycle was calculated by summating stance and swing. Total (i.e., overall leg) positive and negative work during stance, swing, and the whole stride were calculated by summating the values of the six functional muscle groups. The positive and negative work done by each functional muscle group (during stance, swing, and whole stride) was also expressed as a proportion of total leg work done by dividing by the relevant total for all six functional muscle groups. All variables were calculated first for each leg for 6–8 strides at each speed, before averaging across both legs.

sEMG recordings for each speed were analyzed for RMS amplitude throughout five different phases of the gait cycle: braking (from initial contact to horizontal GRF crossing zero), propulsion (from the end of braking to toe-off), early swing (from end of propulsion to initial contact of contralateral leg), contralateral stance (from end of early swing to toe-off of contralateral leg), and late swing (from end of contralateral stance to initial contact of ipsilateral leg). RMS amplitudes during running at each speed were averaged over the 6–8 analyzed strides and over both legs and then normalized to EMG_{CMJ} (see Section 2.4). EMG_{CMJ} was the highest sEMG amplitude, over a 100 ms epoch, during the take-off phase of the three CMJs. Poor quality recordings during the running trials that did not show clear phases of activation or had implausibly large amplitudes were discarded. If recordings from only one leg were affected, signals from the opposite leg were retained and used for the analysis.

2.7 | Statistics

Kinematics and kinetics for one participant at the last speed were discarded due to errors in the motion capture data. The vast majority of the kinematic and kinetic data (95%) were found to be normally distributed using Shapiro–Wilk test, and therefore, parametric statistics were used to determine the effect of running speed. EMG data were not analyzed from two participants due to erroneous signals. One-way repeated measures ANOVA was used to assess the effect of speed. If the assumption of sphericity was violated, values were corrected using the Greenhouse–Geisser correction. Contingent on a main effect of speed with the ANOVA, repeated planned contrasts (PC) were used for post hoc analysis comparing each speed to the preceding speed. Partial eta squared (η^2) was calculated as an effect size as the ratio of the variance explained by the effect of speed divided by the total variance for the ANOVA and the planned contrasts. These statistical analyses were performed with IBM SPSS Statistics Version 21 (IBM Corp., New York, NY, USA). Data are presented as mean ± SD, and differences were considered to be significant if $p < 0.05$. The overall change (Δoverall) was defined as the change between 2.78 and 8.33 m·s^{-1}. Additionally, linear mixed effect models were run in MATLAB to establish relationships between plantar flexor kinetics and stride length, and hip extensor kinetics and stride frequency. The models included fixed and random effects for both intercept and slope with participant as a grouping variable for random effects.

3 | Results

3.1 | Stride Parameters and Ground Reaction Forces

SL and SF increased with each successive speed (effect size (ES) 0.58–0.98; Table 1, Figure 1A). SF increased by 45% and SL by 106% over the range of speeds. The first (2.78–3.89 m·s^{-1}) and second increments (3.89–5.00 m·s^{-1}) in speed were primarily

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TABLE 1 | Stride parameters for six different speeds.

Variable	Speed m·s^{-1}						η^2
	2.78	3.89	5.00	6.11	7.22	8.33	
Step length (m)**	1.02 ± 0.06	1.38 ± 0.09$^{\Delta}$	1.68 ± 0.11$^{\Delta}$	1.88 ± 0.14$^{\Delta}$	2.04 ± 0.15$^{\Delta}$	2.10 ± 0.14$^{\Delta}$	0.99
Step frequency (steps·min^{-1})**	164 ± 10	170 ± 11$^{\Delta}$	180 ± 12$^{\Delta}$	196 ± 16$^{\Delta}$	214 ± 16$^{\Delta}$	238 ± 16$^{\Delta}$	0.97
GCT (s)**	0.228 ± 0.022	0.190 ± 0.014$^{\Delta}$	0.162 ± 0.012$^{\Delta}$	0.144 ± 0.008$^{\Delta}$	0.123 ± 0.009$^{\Delta}$	0.111 ± 0.007$^{\Delta}$	0.97
GCD (m)**	0.624 ± 0.062	0.728 ± 0.054$^{\Delta}$	0.795 ± 0.058$^{\Delta}$	0.864 ± 0.046$^{\Delta}$	0.872 ± 0.067	0.916 ± 0.055$^{\Delta}$	0.94
Flight time (s)**	0.139 ± 0.029	0.164 ± 0.018$^{\Delta}$	0.174 ± 0.017$^{\Delta}$	0.165 ± 0.017$^{\Delta}$	0.159 ± 0.016	0.142 ± 0.014$^{\Delta}$	0.58
Swing time (s)**	0.506 ± 0.047	0.518 ± 0.038	0.509 ± 0.039$^{\Delta}$	0.473 ± 0.041$^{\Delta}$	0.441 ± 0.036$^{\Delta}$	0.395 ± 0.031$^{\Delta}$	0.88

Note: Values are displayed as mean ± SD ($n = 11$). Significant repeated measures ANOVA main effect of speed **($p < 0.01$), $^{\Delta}$Planned contrast, significantly different from the previous speed ($p < 0.05$).
Abbreviations: η^2, partial eta squared (effect size); GCD, ground contact distance; GCT, ground contact time; SD, standard deviation.

due to increases in SL (88% and 76%, respectively), whereas the final speed increment (7.22–8.33 m·s^{-1}) was largely due to higher SF (75%, Figure 1B). GCD increased with almost every speed increment (ES 0.71–0.93), whereas GCT (Δoverall −51%; ES ≥0.84) and swing time (Δoverall −22%; >3.89 m·s^{-1} ES 0.40–0.94) decreased with speed. Peak vertical GRF showed progressively smaller increments of +11.4%, +7.2%, and +2.5% with each successive speed, before plateauing above 6.11 m·s^{-1} (Table 2, Figure S1; ES ≥ 0.50). Flight time and vertical impulse both first increased to a peak at 5.0 m·s^{-1} (+~25% vs. 2.78 m·s^{-1}) and decreased thereafter (ES ≥ 0.49), with similar values at 2.78 and 8.33 m·s^{-1}. Peak propulsion GRF increased with speed (ES ≥ 0.62; Δoverall +200%); peak braking GRF increased up to 7.22 m·s^{-1} (ES = 0.65–0.97; Δoverall +106%).

3.2 | Kinematics

During stance, plantar flexion angle typically increased by ~7° over the range of speeds at both IC (ES ≥ 0.42) and TO (ES ≥ 0.40; Table 3, Figures S2 and S3). The stance knee was increasingly flexed as speed increased at both IC (ES 0.34–0.54; Δoverall +5.6°) and at its peak (ES ≥ 0.40; Δoverall +4.1°). The hip had the largest stance leg range of motion (ROM) increase, +13.7° over all speeds (ES = 0.37–0.89; Δoverall +44%), due primarily to increased hip flexion at IC (ES ≥ 0.64; Δoverall +10.4°), but also increased hip extension at TO (ES = 0.42; Δoverall +3.6°). During stance peak ankle plantar flexion (Δoverall +148%), knee extension (Δoverall +172%) and hip extension (Δoverall +196%) angular velocities had relatively linear increases over the range of speeds (ES ≥ 0.87; Figure 1C), with by far the highest joint velocities during stance at high speeds occurring during ankle plantar flexion.

Peak swing leg knee (Δoverall +47°) and hip (Δoverall +40°) flexion, and consequently knee (Δoverall +44°) and hip (Δoverall +51°) ROM, increased with speed (Table 1, Figure S3; ES ≥ 0.63). Flexion and extension peak angular velocities of all three joints generally showed relatively linear increases, with only peak ankle plantar flexion velocity showing notably sublinear increases (Table 1, Figure 1D; ES ≥ 0.71) and peak hip extension velocity exhibiting the largest increase (Δoverall +296%).

3.3 | Kinetics—Stance Phase

Ankle peak plantar flexion moment increased with each speed, but by progressively smaller increments (ES = 0.44–0.91; Δoverall +74%). Ankle plantar flexion positive and negative work generally showed the same pattern (ES 0.48–0.92; Δoverall +96% and +138%, respectively; Figures 2 and 3). Knee extension peak moment, positive and negative work (ES ≥ 0.40) displayed the same inverted parabola as vertical impulse, peaking at 5.0 m·s^{-1}. Peak hip extension moment (Δoverall +111%) and negative hip flexion work (Δoverall +400%) increased up to 7.22 m·s^{-1} (ES ≥ 0.60), and positive hip extension work continuously increased (ES ≥ 0.51; Δoverall +211%).

During stance, total positive work increased with speed by progressively smaller increments (ES = 0.39–0.92), as did total negative work but only up to 7.22 m·s^{-1}. The ankle plantar flexors

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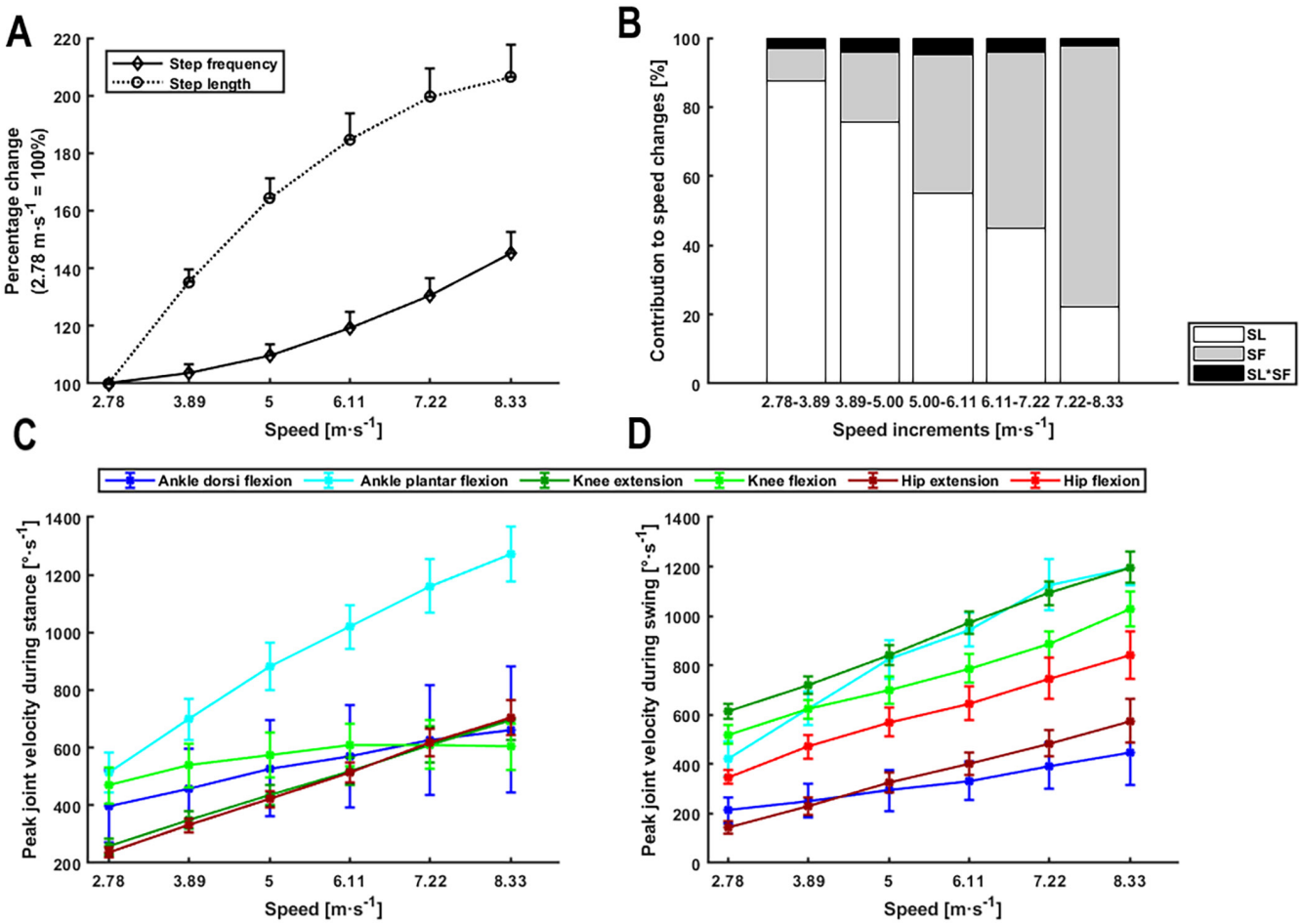

FIGURE 1 | (A) Percentage changes in stride frequency and step length normalized to 2.78 m·s^{-1} at six different running speeds. (B) The contribution of step length, step frequency, and the interaction between both, to each successive increase in speed by 1.11 m·s^{-1}. Peak joint angular velocities of ankle, knee, and hip joints during stance (C) and swing (D) (all shown as positive).

made the largest contribution (>60%) to both total positive and total negative work of the lower limb during stance at all speeds (Figures 2 and 3; Tables S1 and S2). With increasing speed, the knee extensors contributed less to total positive work during stance (overall 17.6%–9.4%; ES ≥ 0.42), whereas the hip extensors contributed more (overall from 15.4% to 24.1%; ES ≥ 0.48). The knee extensors also contributed progressively less to total negative work during stance (overall 27.7%–13.7%; ES = 0.39–0.85), but the hip flexors contributed more (overall 8.2%–17.6%; ES = 0.34–0.75), as speed increased.

3.4 | Kinetics—Swing Phase

During swing, the largest moments were peak hip extension followed by peak hip flexion; both being ~2-fold and 4–7-fold larger than peak knee flexion and extension moments, respectively. Peak knee and hip flexion and extension moments rose superlinearly with speed (ES ≥ 0.66; Δoverall 170%–316%). Similarly, knee negative flexion (Δoverall +472%) and extension work (Δoverall +707%) increased superlinearly with speed (ES ≥ 0.96). Work done at the hip during swing showed superlinear changes with increasing speed (positive and negative hip extension work Δoverall +1130% and +1025%; positive and negative hip flexion work Δoverall +412% and +1133%; ES = 0.41–0.98).

During swing, total positive and negative work increased with progressively larger increments with speed (ES ≥ 0.95). Of the total positive work done by the whole swing leg, there was an increasing hip extensor contribution (overall 36% to 59%; ES ≥ 0.65) and a decreasing hip flexor contribution (overall 46% to 32%; ES ≥ 0.49) with increasing speed. The greatest negative work done during swing was by the knee flexors, but their proportional contribution fell from 62% to 50% (ES = 0.43–0.96) with speed. The contribution to swing negative work by the hip flexors (overall 5.6%–8.9%; ES ≥ 0.35) and hip extensors (overall 7.2%–10.9%; ES ≥ 0.59) and the knee extensors (overall 26.4% to 30.9%; ES = 0.43–0.82) showed modest increases with speed. The contribution of the ankle joint muscles to positive and negative work during swing was negligible.

3.5 | Whole Stride—Sum of Work Done During Unilateral Stance and Swing

Total unilateral positive (overall 1.80–6.61 J·kg^{-1}; $p < 0.001$, ES ≥ 0.94) and negative work (overall −1.72 to −6.65 J·kg^{-1};

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TABLE 2 | Kinetic parameters during running at six different speeds.

Category	Variable	Speed ($m \cdot s^{-1}$) 2.78	3.89	5.00	6.11	7.22	8.33	η^2
Stance								
GRF	Vertical peak GRF ($N \cdot BW^{-1}$)**	2.63 ± 0.22	2.93 ± 0.20$^{\Delta}$	3.14 ± 0.20$^{\Delta}$	3.22 ± 0.19$^{\Delta}$	3.28 ± 0.20	3.28 ± 0.29	0.76
	Vertical impulse ($Ns \cdot kg^{-1}$)**	1.34 ± 0.24	1.59 ± 0.18$^{\Delta}$	1.68 ± 0.19$^{\Delta}$	1.60 ± 0.19$^{\Delta}$	1.52 ± 0.17$^{\Delta}$	1.36 ± 0.16$^{\Delta}$	0.64
	Peak braking GRF ($N \cdot BW^{-1}$)**	−0.31 ± 0.04	−0.41 ± 0.05$^{\Delta}$	−0.50 ± 0.05$^{\Delta}$	−0.57 ± 0.05$^{\Delta}$	−0.62 ± 0.08$^{\Delta}$	−0.64 ± 0.10	0.91
	Peak propulsion GRF ($N \cdot BW^{-1}$)**	0.24 ± 0.04	0.36 ± 0.05$^{\Delta}$	0.49 ± 0.06$^{\Delta}$	0.55 ± 0.06$^{\Delta}$	0.66 ± 0.07$^{\Delta}$	0.72 ± 0.09$^{\Delta}$	0.98
	Horizontal impulse ($Ns \cdot kg^{-1}$)**	0.15 ± 0.04	0.20 ± 0.03$^{\Delta}$	0.23 ± 0.04$^{\Delta}$	0.25 ± 0.04$^{\Delta}$	0.24 ± 0.04	0.23 ± 0.03$^{\Delta}$	0.80
Ankle	Peak plantar flexion moment ($Nm \cdot kg^{-1}$)**	−3.18 ± 0.33	−3.93 ± 0.38$^{\Delta}$	−4.50 ± 0.46$^{\Delta}$	−4.86 ± 0.48$^{\Delta}$	−5.30 ± 0.44$^{\Delta}$	−5.53 ± 0.52$^{\Delta}$	0.95
	Negative plantar flexion work ($J \cdot kg^{-1}$)**	−0.69 ± 0.20	−0.96 ± 0.24$^{\Delta}$	−1.24 ± 0.32$^{\Delta}$	−1.45 ± 0.34$^{\Delta}$	−1.61 ± 0.41$^{\Delta}$	−1.64 ± 0.47	0.90
	Positive plantar flexion work ($J \cdot kg^{-1}$)**	0.80 ± 0.14	1.06 ± 0.16$^{\Delta}$	1.28 ± 0.18$^{\Delta}$	1.43 ± 0.23$^{\Delta}$	1.50 ± 0.27	1.57 ± 0.22$^{\Delta}$	0.92
Knee	Peak extension moment ($Nm \cdot kg^{-1}$)**	2.22 ± 0.48	2.67 ± 0.57$^{\Delta}$	2.72 ± 0.55	2.62 ± 0.54	2.39 ± 0.59$^{\Delta}$	2.25 ± 0.56	0.49
	Negative extension work ($J \cdot kg^{-1}$)**	−0.31 ± 0.10	−0.38 ± 0.08$^{\Delta}$	−0.42 ± 0.10$^{\Delta}$	−0.41 ± 0.12	−0.37 ± 0.12$^{\Delta}$	−0.35 ± 0.11	0.40
	Positive extension work ($J \cdot kg^{-1}$)**	0.22 ± 0.08	0.26 ± 0.07$^{\Delta}$	0.28 ± 0.08	0.28 ± 0.09	0.26 ± 0.08	0.23 ± 0.09$^{\Delta}$	0.35
Hip	Peak extension moment ($Nm \cdot kg^{-1}$)**	−2.19 ± 0.38	−2.90 ± 0.57$^{\Delta}$	−3.37 ± 0.55$^{\Delta}$	−3.92 ± 0.53$^{\Delta}$	−4.35 ± 0.46$^{\Delta}$	−4.63 ± 0.78	0.88
	Positive extension work ($J \cdot kg^{-1}$)**	0.19 ± 0.04	0.26 ± 0.06$^{\Delta}$	0.33 ± 0.05$^{\Delta}$	0.42 ± 0.09$^{\Delta}$	0.50 ± 0.12$^{\Delta}$	0.59 ± 0.14$^{\Delta}$	0.86
	Negative flexion work ($J \cdot kg^{-1}$)**	−0.09 ± 0.04	−0.15 ± 0.06$^{\Delta}$	−0.23 ± 0.08$^{\Delta}$	−0.31 ± 0.12$^{\Delta}$	−0.40 ± 0.16$^{\Delta}$	−0.45 ± 0.22	0.74
Swing								
Knee	Peak extension moment ($Nm \cdot kg^{-1}$)**	0.25 ± 0.06	0.33 ± 0.07$^{\Delta}$	0.44 ± 0.09$^{\Delta}$	0.57 ± 0.13$^{\Delta}$	0.77 ± 0.14$^{\Delta}$	1.04 ± 0.21$^{\Delta}$	0.90
	Peak flexion moment ($Nm \cdot kg^{-1}$)**	−0.78 ± 0.17	−0.96 ± 0.18$^{\Delta}$	−1.20 ± 0.25$^{\Delta}$	−1.45 ± 0.26$^{\Delta}$	−1.78 ± 0.31$^{\Delta}$	−2.21 ± 0.42$^{\Delta}$	0.96
	Negative extension work ($J \cdot kg^{-1}$)**	−0.15 ± 0.03	−0.31 ± 0.06$^{\Delta}$	−0.49 ± 0.09$^{\Delta}$	−0.69 ± 0.12$^{\Delta}$	−0.92 ± 0.15$^{\Delta}$	−1.21 ± 0.18$^{\Delta}$	0.97
	Negative flexion work ($J \cdot kg^{-1}$)**	−0.36 ± 0.06	−0.55 ± 0.08$^{\Delta}$	−0.79 ± 0.11$^{\Delta}$	−1.12 ± 0.16$^{\Delta}$	−1.53 ± 0.20$^{\Delta}$	−2.06 ± 0.28$^{\Delta}$	0.98
Hip	Peak flexion moment ($Nm \cdot kg^{-1}$)**	1.52 ± 0.38	1.83 ± 0.29$^{\Delta}$	2.28 ± 0.32$^{\Delta}$	2.96 ± 0.35$^{\Delta}$	3.55 ± 0.42$^{\Delta}$	4.43 ± 0.52$^{\Delta}$	0.97
	Peak extension moment ($Nm \cdot kg^{-1}$)**	−1.81 ± 0.46	−2.23 ± 0.57$^{\Delta}$	−2.72 ± 0.70$^{\Delta}$	−3.21 ± 0.79$^{\Delta}$	−3.97 ± 0.87$^{\Delta}$	−4.89 ± 1.08$^{\Delta}$	0.93
	Negative flexion work ($J \cdot kg^{-1}$)**	−0.03 ± 0.02	−0.07 ± 0.03$^{\Delta}$	−0.12 ± 0.05$^{\Delta}$	−0.18 ± 0.09	−0.29 ± 0.16$^{\Delta}$	−0.37 ± 0.23$^{\Delta}$	0.71
	Positive flexion work ($J \cdot kg^{-1}$)**	0.26 ± 0.04	0.41 ± 0.07$^{\Delta}$	0.59 ± 0.10$^{\Delta}$	0.79 ± 0.13$^{\Delta}$	1.03 ± 0.20$^{\Delta}$	1.33 ± 0.24$^{\Delta}$	0.95
	Negative extension work ($J \cdot kg^{-1}$)**	−0.04 ± 0.02	−0.09 ± 0.05$^{\Delta}$	−0.16 ± 0.08$^{\Delta}$	−0.23 ± 0.11$^{\Delta}$	−0.33 ± 0.16$^{\Delta}$	−0.45 ± 0.21$^{\Delta}$	0.82
	Positive extension work ($J \cdot kg^{-1}$)**	0.20 ± 0.06	0.42 ± 0.11$^{\Delta}$	0.72 ± 0.15$^{\Delta}$	1.14 ± 0.22$^{\Delta}$	1.68 ± 0.31$^{\Delta}$	2.46± 0.42$^{\Delta}$	0.97

Note: Data are the average of both legs and displayed as mean ± SD ($n = 11$). **Significant repeated measures ANOVA main effect of speed ($p < 0.01$). $^{\Delta}$Planned contrast, significantly different from the previous speed ($p < 0.05$). Abbreviations: η^2, partial eta squared (effect size); BW, body weight; GRF, ground reaction force; SD, standard deviation.

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TABLE 3 | Kinematic parameters during running at six different speeds.

Category	Variable	Speed ($m \cdot s^{-1}$) 2.78	3.89	5.00	6.11	7.22	8.33	η^2
Stance angles								
Ankle	IC (°)**	−0.8 ± 8.4	−1.5 ± 8.5	−3.3 ± 9.5$^{\Delta}$	−5.1 ± 8.8$^{\Delta}$	−6.2 ± 8.2$^{\Delta}$	−7.3 ± 7.4	0.60
	TO (°)**	−10.4 ± 5.0	−12.4 ± 4.8$^{\Delta}$	−12.2 ± 5.4	−15.0 ± 6.6	−13.8 ± 6.1	−17.8 ± 5.8$^{\Delta}$	0.39
	Maximum dorsi flexion (°)**	24.2 ± 3.1	25.5 ± 3.0$^{\Delta}$	26.0 ± 3.3$^{\Delta}$	25.6± 3.5	24.4 ± 3.6$^{\Delta}$	22.5 ± 2.9$^{\Delta}$	0.68
	ROM (°)**	25.0 ± 5.9	27.0 ± 6.4$^{\Delta}$	29.3 ± 7.1$^{\Delta}$	30.7 ± 6.6$^{\Delta}$	30.6 ± 6.3	29.8 ± 6.6	0.56
Knee	IC (°)**	−17.1 ± 2.8	−18.3 ± 2.7$^{\Delta}$	−19.5 ± 2.7$^{\Delta}$	−20.3 ± 2.5	−21.7 ± 2.7$^{\Delta}$	−22.7 ± 3.5$^{\Delta}$	0.66
	TO (°)**	−17.4 ± 4.0	−16.7 ± 3.5	−17.6 ± 3.5	−17.7 ± 3.9	−20.0 ± 4.3$^{\Delta}$	−20.0 ± 4.6	0.41
	Peak flexion (°)**	−41.5 ± 3.2	−43.6 ± 3.1$^{\Delta}$	−44.6 ± 3.1$^{\Delta}$	−45.1 ± 3.6	−45.3 ± 3.7	−45.6 ± 4.3	0.55
	ROM (°)	24.4 ± 2.7	25.3 ± 3.2	25.1 ± 2.6	24.8 ± 2.6	23.6 ± 2.9	22.9 ± 3.0	0.23
Hip	IC (°)**	32.5 ± 3.4	36.7 ± 3.4$^{\Delta}$	40.0 ± 3.7$^{\Delta}$	42.0 ± 4.6$^{\Delta}$	43.30 ± 5.4	42.9 ± 5.2	0.83
	TO (°)*	1.6 ± 4.1	−0.1 ± 4.0$^{\Delta}$	−0.3 ± 4.4	−1.7 ± 5.0	−0.6 ± 5.6	−2.0 ± 5.1	0.29
	ROM (°)**	31.1 ± 2.9	37.1 ± 2.8$^{\Delta}$	40.5 ± 3.0$^{\Delta}$	43.7 ± 2.7$^{\Delta}$	44.0 ± 3.9	44.8 ± 3.7$^{\Delta}$	0.90
Stance velocities								
Ankle	Peak dorsi flexion ($°\cdot s^{-1}$)**	396 ± 131	459 ± 145$^{\Delta}$	529 ± 176$^{\Delta}$	570 ± 186$^{\Delta}$	625 ± 200$^{\Delta}$	662 ± 230	0.84
	Peak plantar flexion ($°\cdot s^{-1}$)**	−514 ± 73	−698 ± 76$^{\Delta}$	−883 ± 87$^{\Delta}$	−1019 ± 80$^{\Delta}$	−1160 ± 98$^{\Delta}$	−1274 ± 99$^{\Delta}$	0.99
Knee	Peak extension ($°\cdot s^{-1}$)**	256 ± 29	350 ± 32$^{\Delta}$	436± 38$^{\Delta}$	517 ± 49$^{\Delta}$	612 ± 65$^{\Delta}$	696 ± 75$^{\Delta}$	0.97
	Peak flexion ($°\cdot s^{-1}$)**	−469 ± 66	−539 ± 79$^{\Delta}$	−576 ± 83$^{\Delta}$	−609 ± 78$^{\Delta}$	−611 ± 89	−604 ± 85	0.64
Hip	Peak extension ($°\cdot s^{-1}$)**	−238 ± 18	−332± 26$^{\Delta}$	−422 ± 31$^{\Delta}$	−514 ± 38$^{\Delta}$	−616 ± 50$^{\Delta}$	−705 ± 64$^{\Delta}$	0.98
Swing angles								
Knee	Peak flexion (°)**	−92.5 ± 8.3	−110.7 ± 9.1$^{\Delta}$	−123.8 ± 8.5$^{\Delta}$	−131.3 ± 9.0$^{\Delta}$	−136.7 ± 8.8$^{\Delta}$	−139.7 ± 9.6$^{\Delta}$	0.96
	ROM (°)**	83.5 ± 9.0	102.1 ± 9.3$^{\Delta}$	114.9 ± 8.9$^{\Delta}$	121.7 ± 9.5$^{\Delta}$	126.6 ± 10.4$^{\Delta}$	127.7 ± 11.8	0.93
Hip	Peak flexion (°)**	43.2 ± 3.1	54.4 ± 4.9$^{\Delta}$	64.8 ± 5.8$^{\Delta}$	72.7 ± 6.9$^{\Delta}$	79.0 ± 7.1$^{\Delta}$	83.8 ± 6.9$^{\Delta}$	0.98
	ROM (°)**	44.5 ± 3.6	60.0 ± 5.3$^{\Delta}$	72.9 ± 5.9$^{\Delta}$	82.6 ± 6.9$^{\Delta}$	90.2 ± 7.8$^{\Delta}$	95.7 ± 8.5$^{\Delta}$	0.99
Swing velocities								
Ankle	Peak dorsi flexion ($°\cdot s^{-1}$)**	213 ± 57	251 ± 72$^{\Delta}$	294 ± 87$^{\Delta}$	332 ± 82$^{\Delta}$	394 ± 100$^{\Delta}$	445 ± 135$^{\Delta}$	0.80
	Peak plantar flexion ($°\cdot s^{-1}$)**	−421 ± 74	−626 ± 71$^{\Delta}$	−824 ± 83$^{\Delta}$	−945 ± 70$^{\Delta}$	−1126 ± 109$^{\Delta}$	−1194 ± 71$^{\Delta}$	0.98

(Continues)

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TABLE 3 | (Continued)

Category	Variable	Speed ($m \cdot s^{-1}$) 2.78	3.89	5.00	6.11	7.22	8.33	η^2
Knee	Peak extension ($°\cdot s^{-1}$)**	614 ± 32	721 ± 35$^{\Delta}$	842 ± 41$^{\Delta}$	973 ± 46$^{\Delta}$	1092 ± 53$^{\Delta}$	1197 ± 65$^{\Delta}$	0.99
	Peak flexion ($°\cdot s^{-1}$)**	−519 ± 39	−623 ± 39$^{\Delta}$	−699 ± 58$^{\Delta}$	−788 ± 61$^{\Delta}$	−888 ± 55$^{\Delta}$	−1028 ± 74$^{\Delta}$	0.99
Hip	Peak extension ($°\cdot s^{-1}$)**	−145 ± 25	−231 ± 36$^{\Delta}$	−324 ± 42$^{\Delta}$	−401 ± 48$^{\Delta}$	−484 ± 55$^{\Delta}$	−576 ± 94$^{\Delta}$	0.96
	Peak flexion ($°\cdot s^{-1}$)**	348 ± 29	471 ± 49$^{\Delta}$	569 ± 62$^{\Delta}$	647 ± 72$^{\Delta}$	747 ± 88$^{\Delta}$	839 ± 101$^{\Delta}$	0.96

Note: Data are displayed as mean ± SD ($n = 11$). Significant repeated measures ANOVA main effect of speed: *($p < 0.05$), **($p < 0.01$). $^{\Delta}$Planned contrast, significantly different from the previous speed ($p < 0.05$).
Abbreviations: η^2, partial eta squared (effect size); IC, initial contact; ROM, range of motion; SD, standard deviation; TO, toe-off.

ES ≥ 0.98) over a stride cycle progressively increased with running speed. The relative contributions of stance and swing phases to the positive and negative work done over a stride reversed as speed increased from ~65%–69% (stance) and ~31%–35% (swing) at 2.78 $m \cdot s^{-1}$, to 38% (stance) and 62% (swing) at 8.33 $m \cdot s^{-1}$. The proportional contribution of the six muscle groups to the positive work of a stride also altered with speed; the plantar flexors were the largest contributor at low speeds, but decreased from 44.7% to 24.2% (ES = 0.50–0.89), whereas hip flexor (overall 14.3% to 20.0%; ES ≥ 0.39) and particularly hip extensor (overall 21.8% to 46.1%; ES = 0.58–0.94) contributions increased with speed, with the latter being the largest contributor to positive work >5 $m \cdot s^{-1}$.

Muscle group contributions to negative work over a stride also changed with speed; knee extensor (overall 27.5%–23.5%; ES ≥ 0.54) and ankle plantar flexor (overall 39.6%–24.6%; ES = 0.41–0.87) contributions decreased as speed increased, but the ankle plantar flexors remained the largest single contributor up to 7.22 $m \cdot s^{-1}$. In contrast, the knee flexors (overall 22.1%–32.2%; ES = 0.65–0.95), hip flexors (overall 7.2%–12.5%; ES = 0.37–0.67), and hip extensors (overall 3.3%–7.0%; ES = 0.38–0.51) all made increasing contributions to whole stride negative work as speed increased.

3.6 | Muscle Activity

During braking and propulsion phases of stance all muscles increased activation with running speed (Figure 4; Figure S4; ES ≥ 0.41). Braking had the highest activation of all the phases for five out of eight muscles (VL, VM, LG, S, and GMax) at all speeds, and for the RF at low speeds, while also being high for the MH and LH (>100% EMG_{CMJ} at 8.33 $m \cdot s^{-1}$). During braking, only the LH and MH continuously increased activation and had the largest changes (Δoverall +321% and +933%, respectively). During propulsion, activation was typically low for knee extensors and GMax across speeds (<25%EMG_{CMJ}), the exceptions being the hamstrings and plantar flexors, with the MH being the only muscle to continuously increase with speed (Δoverall +449%).

During early swing, most muscles had low activation (<10%, or MH <25%) at all speeds, except RF which showed large speed-dependent increases (2.2%–35% EMG_{CMJ} $p \leq 0.041$, ES ≥ 0.53) and had its highest activation of any phase. During contralateral stance (mid-swing), most muscles had low activation at all speeds (≤15% or RF < 25%). During late swing, activation increased superlinearly with speed for the vasti (ES ≥ 0.56), hamstrings (ES = 0.55–0.80), and GMax (ES ≥ 0.86). Moreover, at high speeds activation of the LH and MH during late swing was the highest of any muscles during the stride cycle (~140% EMG_{CMJ}).

3.7 | Relationships Between Kinetic Variables and SL and SF

Peak plantar flexion moment and positive plantar flexion work during stance explained 97% and 96% of the variance in SL with running speed (Figure 5). In addition, peak hip

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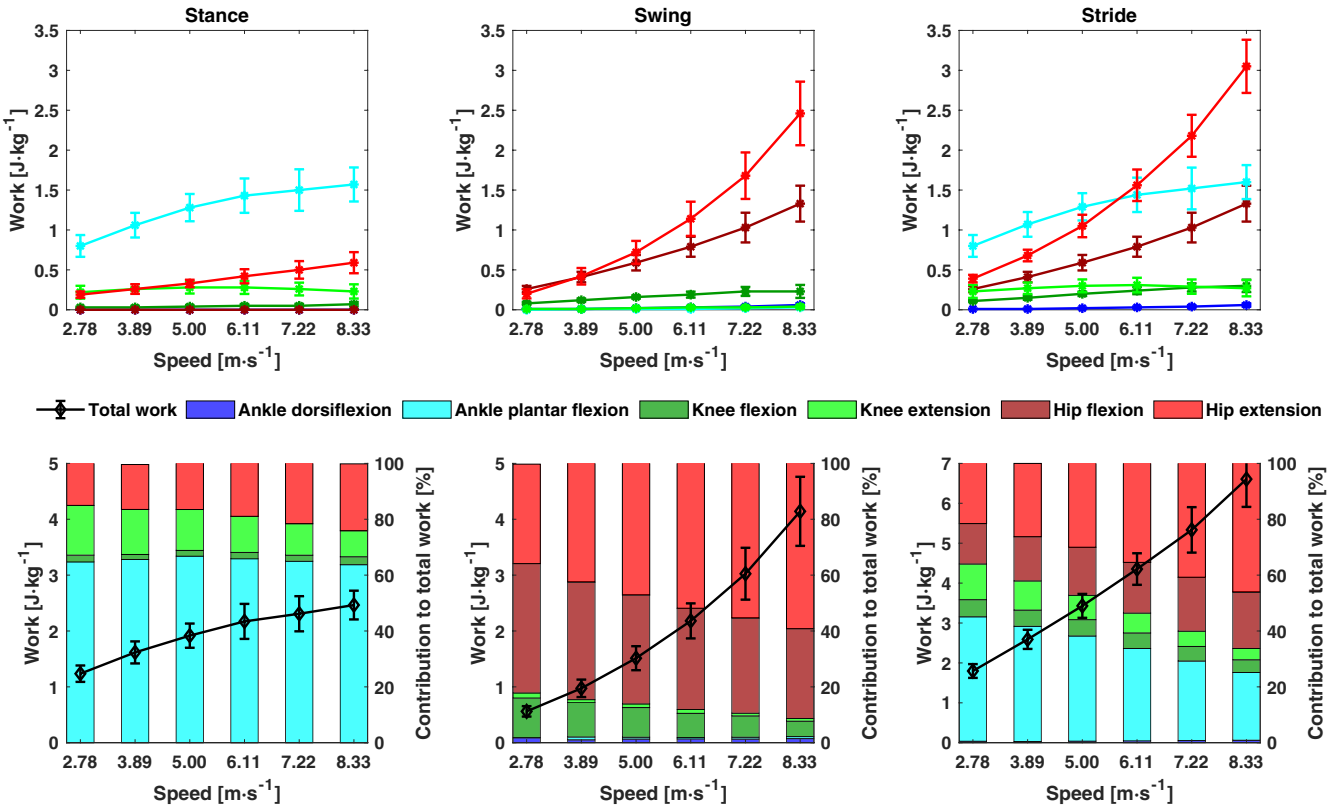

FIGURE 2 | Positive work by flexion and extension moments of the ankle, knee, and hip during stance, swing and overall stride (upper panels), and total positive work of these moments as well as their proportional contributions to total positive work during stance, swing, and the whole stride (lower panels). Data are group mean ± SD ($n = 11$) and unilateral, average of both legs.

extension moment and positive hip extension work during swing explained 98% and 99% of the variance in SF across running speeds.

4 | Discussion

This study compared the neuromechanics of running (kinematics, kinetics, and muscular activation) between a wide range of speeds, and specifically the contribution of the lower limb flexor and extensor muscle groups to the work done during stance, swing, and particularly over a whole stride in a group of runners familiar with running at the full range of speeds. During stance, the ankle plantar flexors produced the highest moments, and positive and negative work (>60% of total positive and negative work) of any muscle group at all speeds, but approached a plateau as speed increased, despite linear increases in plantar flexion activation. As hypothesized, the hip extensors showed a different pattern as speed increased with superlinear increases in positive work and peak moments during swing, and linear increases during stance. Consequently, of the overall positive work done by the legs during the whole running stride (i.e., stance and swing), the hip extensors exceeded the plantar flexors and became the primary contributor (>35%) at >5 m·s^{-1}. Regarding the negative work during swing, the knee flexors (>50%) and extensors (26%–31%) were the major contributors, showing superlinear increases with running speed, such that the knee flexors became the largest contributor to total negative work during the whole stride above 7.22 m·s^{-1}, exceeding the plantar flexors. Hamstrings activation during late swing and braking (early stance) phases also showed the greatest speed-dependent changes in activation of any muscle group and phase. Finally, an original finding was the very strong relationships between plantar flexor moment/positive work during stance with SL, and hip extensor moment/positive work during swing with SF, that highlight the importance of the plantar flexors for increasing SL and the hip extensors for increasing SF.

4.1 | Stride Parameters, Ground Reaction Forces, and Whole Stride Work Contributions

As expected, based on previous work [3], the contribution of SL and SF to the increases in speed was speed-dependent, with SL the major contributor at low speeds and SF the major contributor at higher speeds (>6.11 m·s^{-1}). Despite the progressive decrease in GCT with speed, peak vertical GRF increased at lower speeds before plateauing >6.11 m·s^{-1}. Thus, runners initially compensated for the reduced GCT with an increase in peak vertical GRF until reaching their limit of producing high forces in the short time available. Consequently, vertical impulse showed an inverted "U" shape with increasing speed. In contrast to vertical impulse, GCD increased progressively, mitigating the reduction in GCT and vertical impulse at higher speeds, to help facilitate continued increases in SL. The ability to produce large vertical GRF/impulse, despite decreasing GCT, has been suggested as a key determinant of maximum running speed [2] allowing

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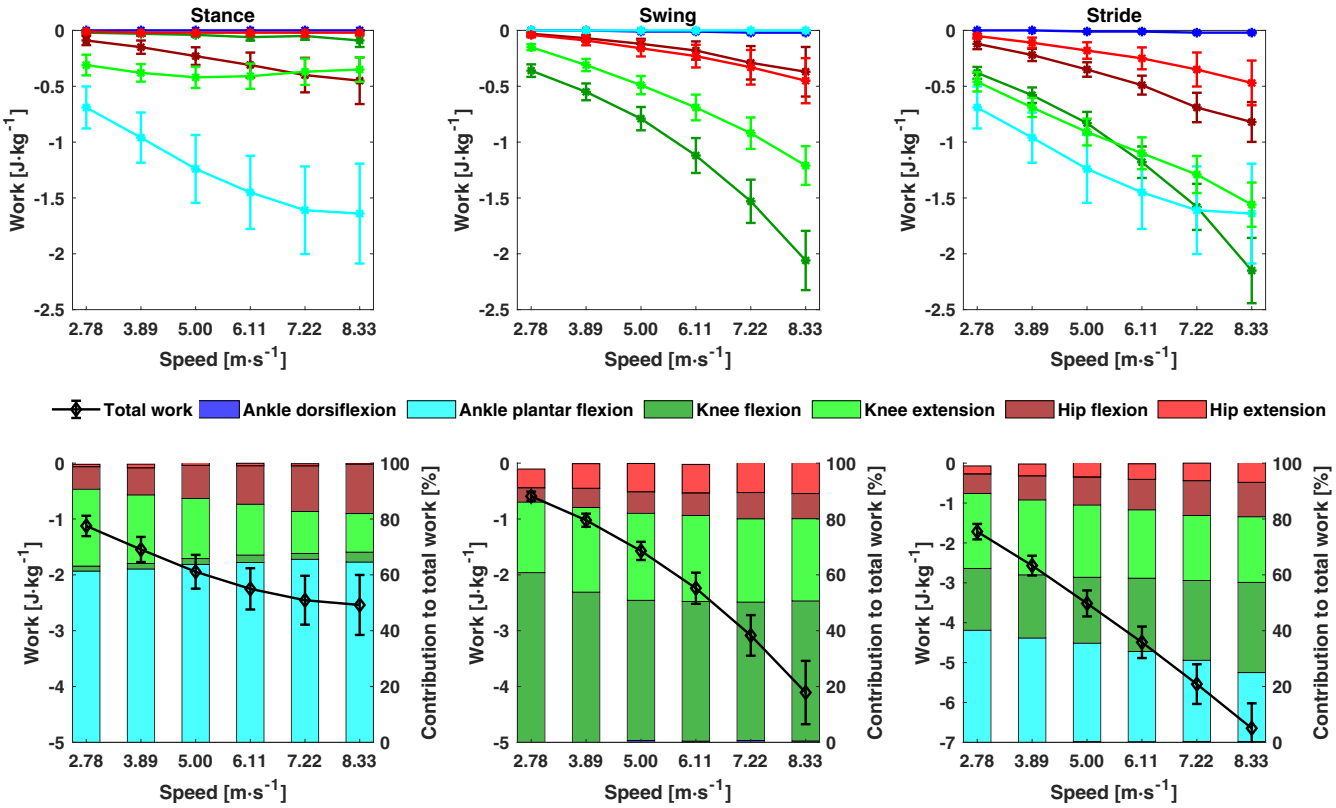

FIGURE 3 | Negative work by flexion and extension moments of the ankle, knee, and hip during stance, swing and overall stride (upper panels), and total negative work of these moments as well as their proportional contributions to total negative work during stance, swing, and the whole stride (lower panels). Data are group mean ± SD ($n = 11$) and unilateral, average of both legs.

sufficient flight time and hence swing time for the leg to be repositioned for the subsequent contact. In contrast to the vertical kinetics, peak propulsion forces increased with every speed increment in accordance with previous research and may support evidence that maximum running speed is related to peak horizontal GRF [30].

For the whole stride, total positive and negative work increased near-linearly with running speed. However, stance and swing phases had differential changes, with stance showing sublinear (as per SL), and swing superlinear (as per SF), increases in both positive and negative total work with running speed. Consequently, for the whole stride there was a progressive shift from stance (~65%–69% at 2.78 m·s^{-1}) to swing (~62% at 8.33 m·s^{-1}) making the major contribution to both positive and negative total work as running speed increased. This increasing reliance on the swing phase to complete the work necessary to run at higher speeds appears to compensate for the inability to achieve linear increases in stance phase work (positive and negative), presumably due to limited contact time.

4.2 | Stance Phase: Plantar Flexor Kinetics and Stride Length

The current study found sublinear increases in peak ankle plantar flexor moments and positive work with increasing running speed, in accordance with previous study [9]. The diminishing increments in plantar flexion kinetics with speed indicate that they were approaching their limit in this cohort at 8.33 m·s^{-1}. This is supported by a previous observation that the plantar flexors produced 96% of their maximum force during sprinting at 9.3 m·s^{-1} [31]. The diminishing increments in plantar flexion kinetics with speed followed a similar pattern to SL. Furthermore, in the current study we directly assessed the relationships between plantar flexor kinetics and SL, finding plantar flexion peak moment and positive work during stance were both very strongly related to SL, explaining 97% and 96% of the variance in SL with running speed, respectively. This relationship between SL and plantar flexor kinetics has previously been inferred from the finding that the plantar flexors were the main contributors to vertical GRF and impulse [3]. However, this inference was not wholly convincing due to the decline in vertical impulse at high speeds, even as SL continued to increase. Nonetheless, the current study provides direct original evidence that plantar flexor kinetics are strongly related to, and may be primarily responsible for, the changes in SL with running speed.

The increments in plantar flexor kinetics up to 8.33 m·s^{-1} were achieved despite increasingly unfavorable mechanical conditions of shorter GCT and faster concentric plantar flexion angular velocity (peak velocity increased by ~2.5-fold), but may have been facilitated by both kinematic and neural changes. Peak dorsiflexion angular velocity during early stance increased approximately twofold with speed. This could have increased the rate at which the plantar flexor muscle-tendon units lengthened,

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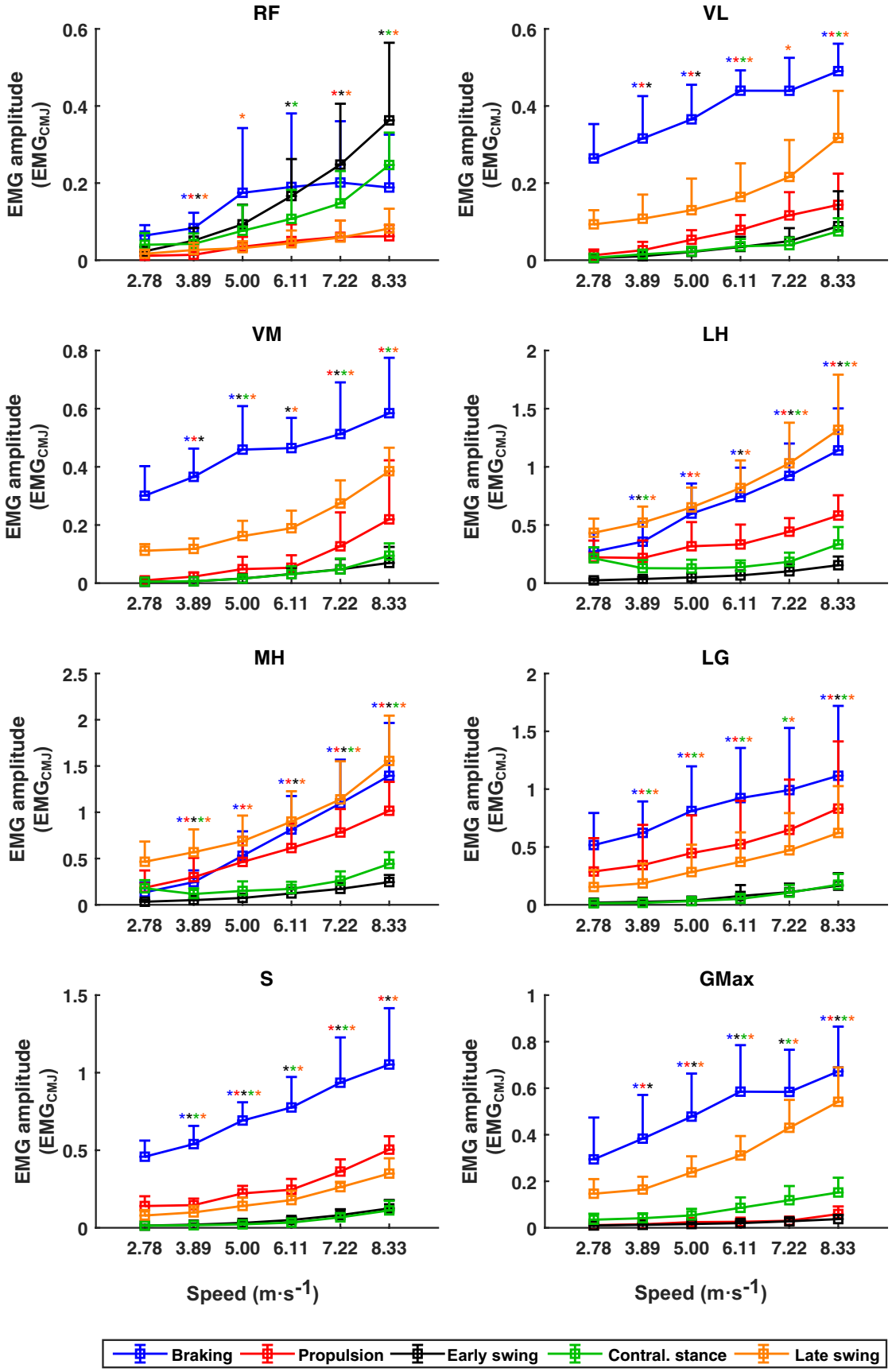

FIGURE 4 | EMG amplitude of eight muscles, average of left and right legs, during five different phases of the running stride, and normalized to the highest RMS amplitude during the take-off of a counter movement jump (CMJ; minimum $n = 7$, maximum $n = 10$). * Significant differences to the previous speed, and colors are based on the different phases.

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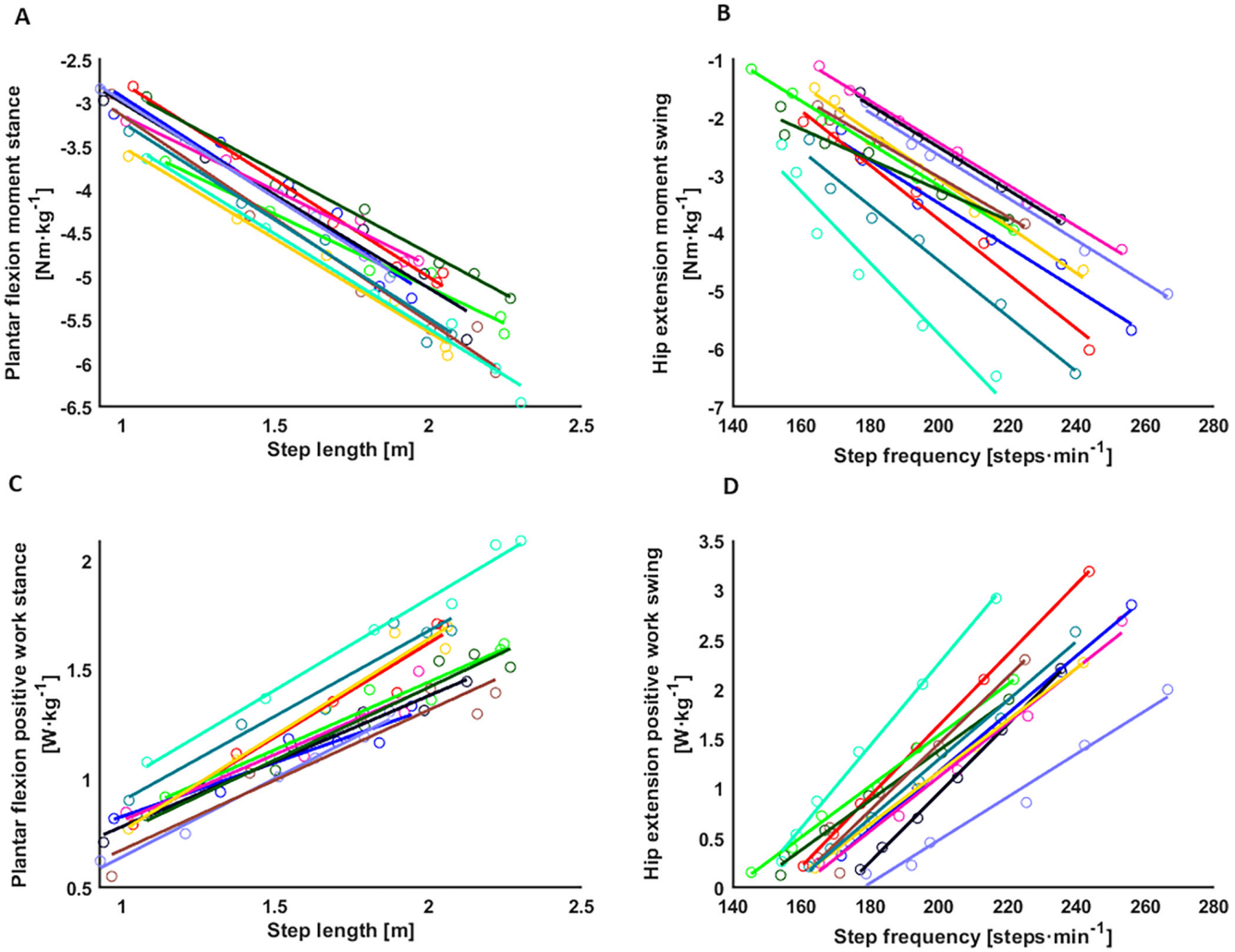

FIGURE 5 | Scatter plots of the relationships between: (A) Peak plantar flexion moment during stance and step length ($R^2 = 0.97$); (B) Plantar flexion positive work during stance and step length ($R^2 = 0.96$); (C) Peak hip extension moment during swing and step frequency ($R^2 = 0.98$); and (D) Hip extension positive work during swing and step frequency from the average of all participants ($R^2 = 0.99$). Individual participant data ($n = 11$) and lines of best fit are shown. Data were analyzed with a linear mixed effect model.

allowing larger forces to be developed and increasing Achilles tendon strain. The tendon recoil could in turn have reduced the rate of shortening of the plantar flexor muscle fascicles during the subsequent concentric contraction, mitigating the effects of the high ankle plantar flexor angular velocity. Furthermore, the increasingly plantarflexed ankle angle at IC and throughout stance at higher speeds (Figure 2) moved the COP backward, potentially decreasing the plantar flexion moment arm which could facilitate higher vertical ground reaction forces for a given plantar flexion torque. Plantar flexor muscle activation during stance (braking and propulsion phases) also increased at least linearly throughout the range of speeds and likely facilitated increased plantar flexor kinetics.

4.3 | Implications for the Plantar Flexors

The contribution of the plantar flexors to total positive and negative work during stance was consistently high at all speeds (>60%) despite being the smallest of the five muscle groups investigated [32]. Hence, the plantar flexors appear to be the main energy generator and absorber during stance irrespective of speed. This is consistent with simulation studies showing the plantar flexors contribute substantially to vertical (~50%) and horizontal (50%–90%) GRF [33] and experimental evidence that the plantar flexors work at a higher capacity than the knee extensors during running [31]. In fact, we recently found that peak planter flexion moment showed the largest decline of the major leg muscle groups during running to exhaustion at 5.7 m·s^{-1} [15], and a similar decline in plantar flexor work done has been found during 10 km running [29]. These large kinetic demands may explain the high incidence of posterior lower leg injuries among long-distance runners [34] and support a greater emphasis on plantar flexor conditioning for both injury prevention and performance enhancement of these athletes.

4.4 | Stance Phase: Knee Extensor and Hip Flexor/Extensor Kinetics

At higher running speeds, the knee was progressively more flexed at IC, as previously reported [20], which was associated with

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superlinear increases in hamstring activation during late swing and braking phases, holding the knee in a more flexed position at IC [20]. Knee extensor peak moment, and positive and negative work all showed an inverted "U" relationship with speed, which may explain the contradictory previous findings of peak knee extension moment increasing [5, 7], not changing [9], and decreasing [11] with speed. These conflicting findings are likely due to the narrow but variable ranges of speeds assessed, perhaps capturing only part of the full relationship. Thus, the current study clarifies that the changes in peak knee extension moments with speed follow to an inverted "U" relationship. We also found the proportional contribution of the knee extensors to total negative and positive work during stance progressively decreased with running speed (both <28% to <14%). Although this contribution is considered important for the support of body weight [35], overall the knee extensors played a relatively minor and diminishing role in the total work done during stance as speed increased.

The speed-related increase in peak hip extension moment during stance is congruent with previous findings [5, 9]. Positive hip extension work also increased substantially with speed (Δoverall ~3-fold), more than other muscle groups, resulting in a greater proportional contribution of the hip extensors to total positive work (Δoverall from 15% to 24%). This could be a result of increased activation of both hamstrings and gluteus maximus during braking and propulsion. We also found an increased contribution of the hip flexors to total negative work during stance (Δoverall from 8% to 18%). Although previous studies have not calculated the direction-specific work done by joint flexors and extensors, our findings are in agreement with a previous analysis of net joint work [13]. Thus, the current study has further elucidated the specific contributions and importance of the hip functional muscle groups to energy generation (extensors) and absorption (flexors) with increases in running speed, even during stance.

4.5 | Swing Phase Neuromechanics and Step Frequency

The increased hip (46°–96°) and knee ROM (84°–128°) during swing, as well as decreased swing time, necessitated higher knee and hip velocities (2- to 4-fold increases), caused by superlinear increases in peak moments. The hip extensors had a major role in generating energy during swing even at low speeds, but exhibited a pronounced speed induced increase (12-fold), that led to this muscle group generating most of the energy during swing at high speeds (from 36% to 59% overall). The actions of the hip extensors during late swing have been suggested to enable increases in SF at high speeds [3], and the current study demonstrated strong relationships between SF and hip extensor kinetics (peak moment and positive work during swing), which explained 98% and 99% of the variance in SF. Thus, hip extensor positive work appears a critical determinant of SF.

The late swing phase of running is known to require negative hip extension and knee flexion work to decelerate the forward rotation of both the thigh and shank prior to ground contact, particularly at high speeds [36]. We found the knee flexors were the main energy absorber during swing at all running speeds with a superlinear increase in negative work (~6-fold). In contrast, perhaps surprisingly, negative hip extension work, even at the fastest speed, was relatively modest (<25% of knee flexion negative work). This is because hip extension moment results in knee extension, transferring angular momentum from the thigh to the shank. Therefore, the predominant negative work done by the hamstrings during the late swing phase was to decelerate the shank, rather than the thigh, as the primary action at the hip during this phase was positive hip extension work. The large increase in positive hip extension work done during late swing at faster running speeds likely results in greater negative momentum of the limb when it contacts the ground, which could augment the subsequent actions of the stance phase muscles. In fact, computer simulations of sprinting have shown that more negative vertical momentum of the leg at touchdown can put the joints of the stance leg in faster eccentric conditions and at superior joint angles, allowing higher joint torques and vertical GRF during the early stance phase [37]. Hamstrings EMG during the late swing phase rose superlinearly with running speed, and at 8.33 $m \cdot s^{-1}$ had the highest normalized EMG of any muscles in this study, reinforcing their very high activation during this phase of sprinting [11, 18], as they simultaneously contributed to both negative knee flexion work and positive hip extension work.

4.6 | Implications for the Hamstring Muscles

At higher running speeds, there was a clear shift away from the plantar flexors to the hip extensors (positive work) and knee flexors (negative work) becoming the biggest contributors to the work done during running. The large contributions of these muscle groups during fast running concur with our recent findings that the volumes of these muscle groups discriminated between elite vs subelite sprinters and were related to sprint performance [32]. The late swing phase of high-speed running involved both large negative/eccentric moments and work by the knee flexors at long muscle lengths (high hip flexion angles), conditions that leads to high mechanical stress within the biarticular hamstrings and appears to explain the high incidence of hamstrings strain injury in athletes competing in sprint/jumps events [34]. Collectively, these findings support the importance of conditioning the hip extensors and knee flexors for enhancing performance and mitigating injury risk during high-speed running events.

4.7 | Limitations

To assess the work done by functional muscle groups, this study calculated the positive and negative work done by each net internal moment (e.g., flexion and extension moment at each joint), on the basis that the net internal moment indicates the predominantly active muscle group. However, due to co-contraction the moment produced by a muscle group is likely to be higher than the net joint moment, which may be considered a minimum possible value. Muscle activation of a wide range of lower limb muscles was measured with surface EMG (16 electrodes); however, due to technical difficulties with some speeds and muscles the sample size was in some isolated cases reduced ($n = 7$). Further other muscles of interest were not or could not be measured with surface EMG, limiting the data interpretation. It should also be noted that surface EMG amplitude measurements may

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be sensitive to cross-talk from other muscles, as well as changes in pre-activation and time shifts in muscle activity across running speeds that could influence the interpretation of EMG amplitude within the five measured phases. All participants completed two sessions on the instrumented treadmill prior to the measurements, and many had prior treadmill running experience. Nonetheless, the highest common speed achieved was $8.33\,m{\cdot}s^{-1}$, even though most runners were likely not at their maximum sprinting speed as they were typically reluctant to run at higher speeds on the treadmill. Although it seems likely that the contributions from both the swing phase of the gait cycle and the nonplantar flexor muscles would have continued to increase with speed, we are conscious that elite sprint running speeds are >2 increments above the highest speed measured in this study. Future studies could do more familiarization with sprint speed treadmill running and/or use higher standard participants.

5 | Conclusions

The changing stride parameters with running speed were explained by alterations in specific joint kinetics, with SL being strongly related to plantar flexor moment during stance ($R^2 = 0.97$) and SF strongly related to positive hip extension work during swing ($R^2 = 0.99$). Despite the importance of the ankle plantar flexors as the major energy generator and absorber during stance at all speeds, we found their contribution to the whole stride energy generation and absorption declined with speed. This was due to a progressive shift away from stance and toward swing contributing the majority of the negative and positive work of the whole stride, and specifically superlinear speed induced increases in both positive (hip extension and flexion) and negative (knee flexion and extension) work performed during swing. This led to the hip extensors being the primary energy generator and knee flexors the primary energy absorber over the whole running stride at sprint speed ($8.33\,m{\cdot}s^{-1}$), due in part to large increases in activation of the knee flexors and hip extensors. These speed, phase (stance/swing), and work (positive/negative) dependent contributions of the different muscle groups have extensive implications for athletic conditioning to enhance performance and prevent injury.

5.1 | Perspective

The contribution of functional muscle groups varied markedly with speed, while also being dependent on phase (stance/swing), and the type of work being performed (positive/negative) with implications for athletic conditioning and injury prevention. Our findings further reinforce the importance of the plantar flexors for running at $\leq 5\,m{\cdot}s^{-1}$, as they made the largest contribution of any muscle group, not just during stance, but also to whole stride positive and negative work (45%–35%), despite their relatively modest size. This is coherent with the plantar flexors exhibiting the greatest decline in function during middle [15] and long-distance events [29] and the high incidence of plantar flexor injuries among distance runners [34]. Large speed-dependent changes meant that at $>8\,m{\cdot}s^{-1}$ the largest contributors to the work of a running stride were the hip extensors (positive work) and knee flexors (negative work), which concurs with the importance of these muscle groups for sprint performance [32] and the high incidence of hamstrings injury in sprint athletes [34]. Collectively, these findings support the importance of long-distance athletes conditioning the plantar flexors, and sprint athletes conditioning the hip extensors and knee flexors for enhancing performance and mitigating injury risk.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available within the article or its Supporting Information.

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